

PERIODIC CLASSIFICATION MCQ

1. Total identified elements worldwide

- a) 92
- b) 118
- c) 63
- d) 56

Correct answer: b

2. Naturally occurring elements number

- a) 118
- b) 92
- c) 26
- d) 14

Correct answer: b

3. Dobereiner grouped elements into

- a) periods
- b) groups
- c) triads
- d) octaves

Correct answer: c

4. Newlands law known

- a) triads
- b) octaves
- c) periodicity
- d) atomic number

Correct answer: b

5. Mendeleev arranged elements by increasing

- a) atomic number
- b) atomic weight
- c) mass number
- d) electron affinity

Correct answer: b

6. Moseley proved properties depend

- a) atomic mass
- b) atomic number
- c) isotopes
- d) valence electrons

Correct answer: b

7. Modern periodic law states properties periodic function

- a) atomic mass
- b) atomic numbers
- c) density
- d) melting point

Correct answer: b

8. Modern periodic table contains

- a) 7 periods 18 groups
- b) 8 periods 16 groups
- c) 6 periods 20 groups
- d) 9 periods 14 groups

Correct answer: a

9. First period contains elements

- a) 1 2

b) 3 10

c) 11 18

d) 19 36

Correct answer: a

10. Longest periods

a) 1st 2nd

b) 4th 5th

c) 6th 7th

d) 2nd 3rd

Correct answer: c

11. Groups vertical columns number

a) 7

b) 18

c) 32

d) 14

Correct answer: b

12. Group 1 elements valence electrons

a) 1

b) 2

c) 7

d) 8

Correct answer: a

13. Noble gases group number

a) 1

b) 2

c) 17

d) 18

Correct answer: d

14.s-block contains groups

a) 1 2

b) 13 18

c) 3 12

d) 3 only

Correct answer: a

15.p-block groups range

a) 1 2

b) 3 12

c) 13 18

d) bottom row

Correct answer: c

16.d-block called

a) alkali metals

b) transition elements

c) noble gases

d) lanthanides

Correct answer: b

17.f-block inner transition elements

a) s p blocks

b) lanthanides actinides

c) groups 1 2

d) periods 1 3

Correct answer: b

18. Atomic radius decreases across period left

- a) right
- b) top
- c) bottom
- d) diagonally

Correct answer: a

19. Atomic radius increases down

- a) period
- b) group
- c) block
- d) series

Correct answer: b

20. Metallic radius half distance adjacent

- a) nonmetal atoms
- b) metal atoms
- c) ions
- d) molecules

Correct answer: b

21. Covalent radius nonmetallic elements

- a) metallic bond
- b) covalent bond
- c) ionic bond
- d) hydrogen bond

Correct answer: b

22. Cation size compared neutral atom

- a) larger

b) smaller

c) equal

d) double

Correct answer: b

23. Anion size compared neutral atom

a) smaller

b) larger

c) equal

d) half

Correct answer: b

24. Ionisation energy increases across

a) group

b) period left right

c) down group

d) diagonal

Correct answer: b

25. Ionisation energy decreases down

a) period

b) group

c) block

d) series

Correct answer: b

26. Electron affinity energy released adding electron

a) cation

b) neutral atom anion

c) anion

d) molecule

Correct answer: b

27. Noble gases electron affinity

a) high

b) zero

c) negative

d) positive

Correct answer: b

28. Electronegativity increases across

a) group

b) period left right

c) down group

d) s-block

Correct answer: b

29. Pauling scale measures

a) atomic radius

b) electronegativity

c) ionisation energy

d) density

Correct answer: b

30. Electronegativity difference 1.7 bond

a) purely ionic

b) purely covalent

c) 50% each

d) metallic

Correct answer: c

31. Minerals metal compounds

- a) pure metals
- b) soil impurities
- c) ores gangue
- d) pure elements

Correct answer: c

32. Ores economically extractable

- a) all minerals
- b) some minerals
- c) nonmetals
- d) rocks

Correct answer: b

33. Gangue rocky impurity

- a) ore
- b) flux
- c) slag
- d) metal

Correct answer: a

34. Flux reduces fusion temperature reacts

- a) ore
- b) gangue slag
- c) metal
- d) coke

Correct answer: b

35. Smelting reduces oxide

- a) slag

b) molten metal

c) flux

d) gangue

Correct answer: b

36.Gravity separation oxide

a) sulphide ores

b) haematite

c) tinstone

d) bauxite

Correct answer: b

37.Magnetic separation

a) haematite

b) tinstone

c) zinc blende

d) bauxite

Correct answer: b

38.Froth flotation sulphide

a) oxide ores

b) zinc blende

c) haematite

d) tinstone

Correct answer: b

39.Leaching pure ore

a) haematite

b) bauxite

c) copper pyrite

d) iron pyrite

Correct answer: b

40. Tamil Nadu titanium minerals

a) Coimbatore

b) Kanyakumari

c) Salem

d) Madurai

Correct answer: b

41. Metals physical property hammered sheets

a) ductility

b) malleability

c) conductivity

d) sonority

Correct answer: b

42. Metals drawn wires

a) malleability

b) ductility

c) hardness

d) lustre

Correct answer: b

43. Metals good conductors heat electricity

a) lustre

b) sonority

c) conductivity

d) tensile strength

Correct answer: c

44. Metals shiny appearance

- a) conductivity
- b) lustre
- c) sonority
- d) hardness

Correct answer: b

45. Metals produce ringing sound

- a) sonorous
- b) malleable
- c) ductile
- d) hard

Correct answer: a

46. Metals valence electrons outermost shell

- a) 5 6 7
- b) 1 2 3
- c) 4 5 6
- d) 7 8

Correct answer: b

47. Metals form positive ions

- a) electronegative
- b) electropositive
- c) neutral
- d) negative

Correct answer: b

48. Aluminium chief ore

- a) haematite

- b) bauxite
- c) copper pyrite
- d) magnetite

Correct answer: b

49. Baeyer process converts bauxite

- a) metal
- b) alumina
- c) cryolite
- d) slag

Correct answer: b

50. Hall process electrolytic reduction

- a) copper
- b) iron
- c) alumina aluminium
- d) haematite

Correct answer: c

51. Aluminium extraction electrolyte

- a) NaOH
- b) cryolite fluorspar
- c) HCl
- d) H₂SO₄

Correct answer: b

52. Aluminium density

- a) 2.7
- b) 8.9
- c) 7.8

d) 10.5

Correct answer: a

53. Aluminium melting point

a) 660°C

b) 1356°C

c) 1538°C

d) 1083°C

Correct answer: a

54. Aluminothermy reduces

a) copper oxide

b) iron oxide

c) aluminium oxide

d) zinc oxide

Correct answer: b

55. Copper chief ore

a) bauxite

b) haematite

c) copper pyrite

d) magnetite

Correct answer: c

56. Copper extraction froth flotation roasting

a) smelting bessemerisation

b) leaching

c) gravity separation

d) magnetic separation

Correct answer: a

57. Blister copper contains impurities

- a) 2%
- b) 98%
- c) 50%
- d) 76%

Correct answer: a

58. Copper density

- a) 2.7
- b) 8.9
- c) 7.8
- d) 11.3

Correct answer: b

59. Copper melting point

- a) 660°C
- b) 1083°C
- c) 1356°C
- d) 1538°C

Correct answer: b

60. Green coating copper moist air

- a) CuO
- b) $\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$
- c) Cu_2O
- d) CuCl_2

Correct answer: b

61. Iron chief ore

- a) copper pyrite

b) haematite

c) bauxite

d) cryolite

Correct answer: b

62. Blast furnace regions three

a) combustion fusion reduction

b) roasting smelting refining

c) leaching flotation gravity

d) magnetic hydraulic chemical

Correct answer: a

63. Iron density

a) 2.7

b) 7.8

c) 8.9

d) 11.3

Correct answer: b

64. Rust chemical formula

a) FeO

b) $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$

c) Fe_3O_4

d) FeCl_3

Correct answer: b

65. Pig iron carbon content

a) $<0.25\%$

b) $0.25-2\%$

c) $2-4.5\%$

d) >4.5%

Correct answer: c

66. Steel carbon percentage

a) <0.25%

b) 0.25-2%

c) 2-4.5%

d) >4.5%

Correct answer: b

67. Wrought iron carbon

a) <0.25%

b) 0.25-2%

c) 2-4.5%

d) >4.5%

Correct answer: a

68. Alloy homogenous mixture

a) nonmetals only

b) metals nonmetal

c) metalloids

d) pure metals

Correct answer: b

69. Brass composition

a) Cu Zn

b) Fe Cr

c) Al Cu

d) Ni Cr

Correct answer: a

70. Stainless steel contains

- a) Cu Zn
- b) Fe Ni Cr
- c) Al Mg
- d) Pb Sn

Correct answer: b

71. Corrosion metal converts stable

- a) compound
- b) alloy
- c) ore
- d) slag

Correct answer: a

72. Dry corrosion absence

- a) water
- b) oxygen
- c) air
- d) heat

Correct answer: a

73. Wet corrosion presence

- a) dry air
- b) moisture
- c) high temperature
- d) vacuum

Correct answer: b

74. Galvanization coating

- a) tin

- b) zinc iron
- c) nickel
- d) chromium

Correct answer: b

75. Anodizing commonly

- a) iron
- b) copper
- c) aluminium
- d) zinc

Correct answer: c

76. Cathodic protection sacrificial

- a) cathode
- b) anode metal
- c) electrolyte
- d) coating

Correct answer: b

77. Pamban Bridge corrosion prevention

- a) alloying
- b) protective coating
- c) galvanization
- d) electroplating

Correct answer: b

78. Aristotle elements

- a) 118
- b) 92
- c) 4 earth water fire air

d) 63

Correct answer: c

79. Lavoisier classified

a) atomic number

b) gases metals nonmetals earths

c) triads

d) octaves

Correct answer: b

80. Fourth period transition elements

a) 8

b) 10

c) 14

d) 18

Correct answer: b

81. Sixth period lanthanides

a) 8

b) 10

c) 14

d) 18

Correct answer: c

82. Valence electrons determine

a) atomic radius

b) chemical properties

c) physical properties

d) density

Correct answer: b

83. Effective nuclear charge increases across

- a) group
- b) period
- c) down group
- d) s-block

Correct answer: b

84. Principal quantum number increases

- a) across period
- b) down group
- c) left right
- d) p-block

Correct answer: b

85. Bayer process temperature

- a) 1000°C
- b) 150°C
- c) 950°C
- d) 660°C

Correct answer: b

86. Hall process temperature

- a) 150°C
- b) 660°C
- c) 900-950°C
- d) 1356°C

Correct answer: c

87. Copper refining electrolytic

- a) cathode pure copper

b) anode pure copper

c) electrolyte NaOH

d) voltage 2V

Correct answer: a

88. Anode mud contains

a) pure copper

b) impurities

c) slag

d) flux

Correct answer: b

89. Ferrous alloys contain iron

a) major component

b) minor component

c) no iron

d) equal iron

Correct answer: a

90. Wood metal powdered alloy

a) Cu Zn

b) Pb Sn Bi Cd

c) Fe Cr Ni

d) Al Mg

Correct answer: b

91. Corrosion prevention alloying example

a) pure iron

b) stainless steel

c) pure copper

d) pure aluminium

Correct answer: b

92. Electroplating deposits

a) base metal

b) coating metal

c) alloy

d) slag

Correct answer: b

93. Haematite formula

a) FeS_2

b) Fe_2O_3

c) Fe_3O_4

d) FeCO_3

Correct answer: b

94. Copper pyrite formula

a) CuFeS_2

b) Cu_2S

c) Cu_2O

d) CuCO_3

Correct answer: a

95. Ionic radius cation charge increases

a) increases

b) decreases

c) same

d) doubles

Correct answer: b

96. Electron gain enthalpy noble gases

- a) high positive
- b) zero
- c) high negative
- d) low positive

Correct answer: b

97. Metallurgy three main steps

- a) mining extraction refining
- b) concentration production refining
- c) roasting smelting bessemerisation
- d) leaching flotation gravity

Correct answer: b

98. Flux gangue produce

- a) metal
- b) ore
- c) slag
- d) matte

Correct answer: c

99. Blast furnace charge ratio ore coke limestone

- a) 8:4:1
- b) 1:1:1
- c) 4:8:1
- d) 1:4:8

Correct answer: a

100. Iron passivity concentrated

- a) HCl

b) H_2SO_4

c) HNO_3

d) NaOH

Correct answer: c